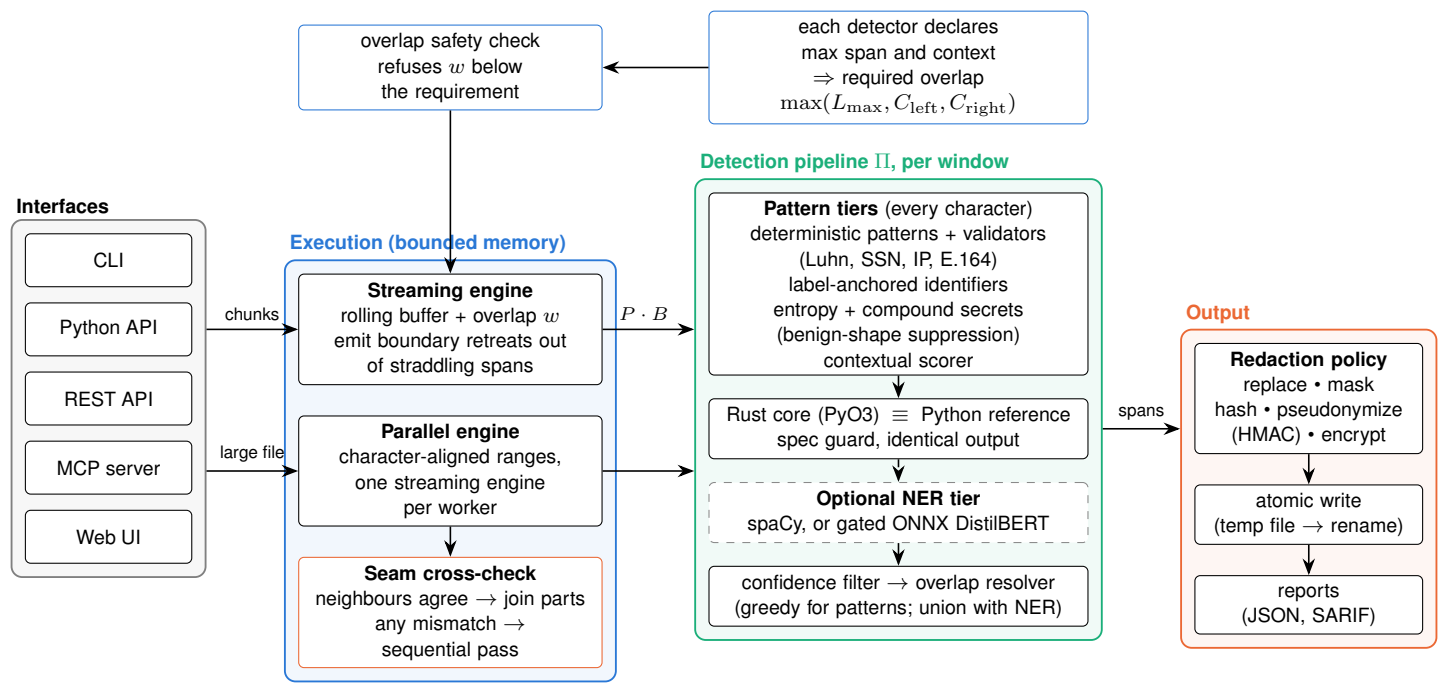

each detector declares
max span and context
⇒ required overlap
$\max(L_{\max}, C_{\text{left}}, C_{\text{right}})$
overlap safety check
refuses w below
the requirement
Interfaces
CLI
Python API
REST API
MCP server
Web UI
Execution (bounded memory)
chunks
large file
Streaming engine
rolling buffer + overlap w
emit boundary retreats out
of straddling spans
Parallel engine
character-aligned ranges,
one streaming engine
per worker
Seam cross-check
neighbours agree → join parts
any mismatch →
sequential pass
$P \cdot B$
Detection pipeline Π, per window
Pattern tiers (every character)
deterministic patterns + validators
(Luhn, SSN, IP, E.164)
label-anchored identifiers
entropy + compound secrets
(benign-shape suppression)
contextual scorer
Rust core (PyO3) ≡ Python reference
spec guard, identical output
Optional NER tier
spaCy, or gated ONNX DistilBERT
confidence filter → overlap resolver
(greedy for patterns; union with NER)
spans
Output
Redaction policy
replace • mask
hash • pseudonymize
(HMAC) • encrypt
atomic write
(temp file → rename)
reports
(JSON, SARIF)